

Imagination and False Memory Inductions: Investigating the Role of Process, Content and Source of Imaginations

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SUMMARY

Three experiments examined the influence of imaginal encoding on false memory rates induced by the Deese-Roediger-McDermott (DRM) paradigm. Several features of imaginal encoding were investigated including the processes giving rise to the images (automatic or deliberate), the source of imagery scripts (participant or another person), and imagery content (e.g. individual or integrated images). Deliberately generating images led to a reduction in false memory rates relative to an imagery control condition. However, the effects on false memory rates very much depended on features characterizing the imaginal encoding. Most notably, when participants (rather than someone else) specified the scripts for guiding their imagery experiences, false memory rates were considerably less. Extending the generality of the distinctiveness heuristic as an explanation for the reduction in false memory rates following pictorial encoding, the results also speak to debates about the wisdom of using guided imagery techniques in applied settings. Copyright © 2006 John Wiley & Sons, Ltd.

Imagination's powerful enhancing effects on learning, remembering, and decision-making have long been recognized. Memory enhancement for both laboratory (Foley, Wilder, McCall, & Van Vorst, 1993; McDaniel & Pressley, 1987) and educational materials (Pressley & Brewster, 1990) follows from the use of imaginal encoding strategies. Similarly, imagining actions in story lines enhances reading comprehension (Glenberg, Gutierrez, Levin, Japuntich, & Kaschak, 2004) by way of facilitating inferential processes about intentions and goals (Turner, 1996). Moreover, the evocative power of literary works to transport readers by way of imagination 'into' others' lives (e.g. Green, Strange, & Brock, 2002; Spolsky, 1997) is thought to contribute significantly to a wide range of ethical decision-making (Johnson, 1993; Nussbaum, 1995, 2003). Nevertheless, these same rich qualities of imaginal processing have their downside, on occasion leading to considerable memory confusions.

The negative effects of imaginal processing on remembering are well documented, making all the more compelling concerns about their downside in both legal (Brainerd & Reyna, 2005) and clinical settings (Arbuthnott, Arbuthnott, & Rossiter, 2001; Brown, Schefflin, & Hammond, 1998; Madill & Holch, 2004). In the experiments reported in this manuscript, we test predictions about the effects of imaginal encoding on memory confusions. With the goal of advancing our understanding of the mechanisms contributing

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to false memories prompted by imaginal encoding, these tests should also inform debates about the use of guided imagery techniques in applied settings.

After responding to explicit requests to generate images of discrete actions (Foley & Ratner, 1998) or more integrated action sequences (Hashtroudi, Johnson, & Chrosniak, 1990), both adults (Hashtroudi et al., 1990) and children (e.g. Foley & Ratner, 1998) later experience confusion when asked to remember what they did and what they only imagined doing. These memory confusions are exacerbated by requests to repeatedly imagine objects (e.g. Johnson, Raye, Wang, & Taylor, 1979), discrete actions (e.g. Goff & Roediger, 1998), or extended episodes (e.g. Garry, Manning, Loftus, & Sherman, 1996; Garry & Wade, 2005), leading to significant memory errors in a variety of contexts.

Memory confusions derived from imaginal processing also occur in the absence of explicit requests to generate images (e.g. Foley, Durso, Wilder, & Friedman, 1991; Read, 1996). These spontaneous images are easily induced, and often lead to confusions between memories for perceptions and memories for thoughts. Subsequent to seeing familiar objects embedded in complex scenes, adults report 'seeing' related objects never actually visible in those scenes, whether experienced live (Lampinen, Copeland, & Neuschatz, 2001), pictorially (Malmstrom & LaVoie, 2002), or visually incomplete (Foley, Foley, Durso, & Smith, 1997).

Although the processes giving rise to imagery experiences play a role in creating memory confusions, as yet we know little about the features of imaginal encoding that contribute to memory confusions under less directive conditions. By less directive conditions, we refer to situations in which images are experienced spontaneously as well as those in which explicit requests to generate images do not include descriptions for the content of those images. In the present series of studies, we began to explore the effects of imaginal processing under these less directive conditions by using the most heavily investigated procedure for studying false memory errors, namely the Deese-Roediger-McDermott, or DRM, procedure (Deese, 1959; Roediger & McDermott, 1995).

In a prototypical version of the DRM procedure, after listening to an experimenter read lists of thematically related words (e.g. table, couch, lamp, desk), participants falsely report that other words, thematically related to those presented (e.g. chair) but not actually presented themselves, were included in the encoding series. Compared to baseline levels of false alarms (e.g. 20–30%, Underwood, 1965), false memories accompanying the presentation of related word lists are often quite high ranging from 70–80% (e.g. Roediger & McDermott, 1995; Roediger, McDermott, & Robinson, 1998). False memories in the DRM procedure are observed across a range of encoding conditions including those involving implicit tasks (Hicks & Starns, 2005) and across a range of test conditions (e.g. Brainerd & Reyna, 2005; Roediger & McDermott, 1995; for a review, see Roediger et al., 1998).

Despite the robustness of false memory effects created by the DRM paradigm, changes in the modality in which items are presented can lead to substantial reductions in these errors (Dodson & Schacter, 2001; Gallo, McDermott, Percer, & Roediger, 2001; Gallo & Roediger, 2003; Kellogg, 2001; Smith & Hunt, 1998). The modality effects of particular interest to us are those following the pictorial presentation of target items. When items are presented as pictures rather than as words, false memory rates decrease considerably (e.g. from 0.68 to 0.30, Israel & Schacter, 1997, Experiment 2). These pictorial modality effects are provocative in their suggestion that similar reduction effects in false memories may follow imaginal encoding.

False memory errors induced by the DRM procedure are thought to reflect both the influence of implicit associative responses activated during encoding (Hege & Dodson,

2004; Roediger & McDermott, 1995; Underwood, 1965) and the failure at test to monitor the source of the implicit responses (Dehon & Bredart, 2004; Gallo, 2004; Dodson & Schacter, 2001; Johnson, Hashtroudi, & Lindsay, 1993; Rotello & Heit, 1999). With the presentation of related items, additional words come to mind by way of the spreading of activation of associative networks. When encountering these implicit responses as test items, participants must correctly monitor their source, remembering that these items were experienced only in thought (Johnson et al., 1993). Although the particular mechanisms(s) invoked to account for monitoring failures differ, a number of theoretical frameworks emphasize the contribution of these monitoring failures to the creation of false memories in the DRM paradigm (e.g. Brainerd & Reyna, 2005; Dodson & Schacter, 2001; Gallo, 2004; McDermott & Watson, 2001).

Both the activation of implicit associative responses and monitoring processes activated at test are invoked to explain the effects of pictorial encoding. What interests us about these explanations (referred to here as the *impoverished relational encoding* hypothesis and the *distinctiveness heuristic*, respectively) is their emphasis on the possible role of imaginal processing. According to the impoverished relational-encoding view, with the presentation of pictures, participants may be more likely to focus on features of these pictures or, they may be more likely to re-experience specific details at test by way of visualizing the pictures experienced during encoding (Hege & Dodson, 2004). In either case, the pictures, and their corresponding internal representations, are thought to reduce (or preclude) the relational processing that is thought to support implicit activation processes.

The *distinctiveness heuristic* shifts attention from encoding to retrieval processes, but again, implies that imaginal processing plays a role in the reduction of false memories following pictorial presentation (Dodson & Schacter, 2002a,b, 2001; Gallo, 2004; Johnson et al., 1993; Johnson & Raye, 2000; Schacter, Israel, & Racine, 1999). A monitoring strategy that draws on distinctiveness as a diagnostic could be influenced by beliefs about the way distinctiveness influences memory, by global impressions of distinctiveness associated with a set of presented items, or by impressions of distinctiveness associated with individual items themselves.

A pictorial presentation mode may set up the expectation that presented items should be relatively memorable. Thus, test items not experienced as such will be rejected as having been experienced. Pictorial presentations may also create a global impression of list distinctiveness such that when new items (including critical lures) are presented at test, the absence of impressions of distinctiveness may signal that the items were not part of the lists presented previously (Dodson & Schacter, 2001; Gallo, 2004; Schacter et al., 1999). Finally, the presentation of test items could lead to the re-activation of specific images based on the actual pictures viewed during encoding, again, facilitating memory decisions. If a test item does not evoke a specific image, this 'absence' may be interpreted as evidence that the item was not presented earlier. Thus, when a critical lure such as *chair* is presented at test, even if the item feels familiar, presumably the item will be rejected as presented previously because a specific image corresponding to a picture included in the encoding series does not come readily to mind at test.

Explanations for pictorial encoding effects invite the test of predictions about the influence of imaginal processing on false memory rates. Surprisingly, investigations of the effects of imaginal processing on false memory rates in the DRM procedure are scant (Kellogg, 2001; Newstead & Newstead, 1998). Nonetheless, we focused on this procedure for a number of reasons. Perhaps in part because memory illusions induced by the DRM paradigm appear to be much stronger than those induced by other kinds of paradigms

(Marsh & Bower, 2004), these particular kinds of false recollection are discussed in ways that serve to fuel debates about the implications of false memory research for applied settings (e.g. Bruce & Winograd, 1998; DePrince, Allard, Oh, & Freyd, 2004; Freyd & Gleaves, 1996; Roediger & McDermott, 1998). Roediger and his colleagues have suggested that the kinds of false memories produced by the DRM paradigm may not be unlike those induced by therapists (or well-intentioned significant others) when engaged in conversations about the past (e.g. Roediger & McDermott, 1998). Not surprisingly, then, false memory effects including those induced by the DRM procedure are also frequently used to buttress concerns about the use of guided imagery techniques in applied settings, calling into question the wisdom of their use (Ceci & Loftus, 1994; Loftus, 2004). Therefore, the extent to which imaginal encoding, including that involving guided imagery, induces false memories in the DRM paradigm deserves attention for both theoretical and applied reasons.

In much the same way that pictures may be re-experienced as images at test, leading to reductions in false memory rates, images experienced during encoding may be re-experienced at test, leading to similar reductions. One study of the effects of orthographic imagery in the DRM paradigm is at least suggestive in this regard. Kellogg (2001) reported that orthographic imagery accompanying the auditory presentation of words reduced false memory recall. Moreover, our previous research on memory confusions investigated in other contexts also supports this line of reasoning. After describing functions for objects presented as pictures or words, adults and children later exhibit considerable confusion, reporting that they saw pictures for objects only presented as words (Durso & Johnson, 1980; Foley et al., 1991). However, in response to explicit requests, if adults or children generate images of objects presented as pictures or words, when later asked to remember whether objects were experienced as pictures or words, both adult and child age groups are quite good at remembering presentation modality.

Thought to reflect the nature of the processes giving rise to the imagery experiences, these picture/word confusions are predicted by the Source-Monitoring framework (Durso & Johnson, 1980; Foley et al., 1991; Johnson et al., 1993; Johnson & Raye, 2000). Presumably the presentation of words referring to familiar objects (or the presentation of the objects themselves) implicitly activates images of those objects. When later asked to remember what objects were actually seen as (or embedded in) pictures, recollections of these implicit images may then be misremembered as seen (Foley et al., 1991; Intraub & Hoffman, 1992; Johnson et al., 1993). However, when images are based on more deliberative processes (in response to requests to generate images), cues related to these generative acts should serve as a diagnostic for picture/word judgements, thereby reducing (or eliminating) memory confusions.

These findings led us to expect that, in the DRM paradigm, if participants are asked to generate images of items, recollecting these images at test should help distinguish presented from non-presented related items. Even if images corresponding to lures (e.g. chair) are evoked spontaneously during encoding, and are re-experienced at test, participants should still be able to distinguish presented from non-presented items. In this case, at test, participants should be able to draw on cues about the processes mediating the imagery experiences—deliberate in the case of presented items, and spontaneous in the case of non-presented related ones.

In other contexts, requests to generate images of individual objects spontaneously evoke elaborative processing such as the visualization of complex scenes or conversational episodes, leading to memory enhancement effects (Foley, Belch, Mann, & McLean, 1999)

including reductions in memory confusions about what was said or only experienced in thought (Foley, Santini, & Sopasakis, 1989). These spontaneous elaborations are often self-referential in nature with participants reporting thinking about objects, persons or episodes from their own lives (Foley et al., 1999). If similar kinds of elaborations accompany requests to generate images in the present studies, then we would expect reductions in false memory rates.

However, other lines of research lead to very different predictions about the effects of imaginal encoding on false memory rates. Investigations of the imagination inflation effect demonstrate that simply asking participants to generate images of actual or plausible events can lead to considerable memory confusion, with adults reporting they experienced events more often than they did or reporting they may have experienced events that never actually occurred (e.g. Garry et al., 1996). Moreover, using the DRM paradigm itself, Marsh and Bower (2004) demonstrated that asking participants to describe what comes to mind during encoding—a form of generating—subsequently led to an inflation of the false memory effect. From this perspective, asking participants to generate images may lead to an increase in false memory errors.

With the goal of testing these competing predictions about the possible influence of imaginal encoding on false memory rates, the experiments reported in this manuscript focused on three features of imaginal encoding, namely, the processes giving rise to imagery experiences (automatic vs. deliberate), the content of those imagery experiences, and the source of the descriptions on which the imagery experiences were based (participant generated vs. scripted). These features were examined by way of various instructional manipulations. To examine the role of imaginal processing, false memory rates following explicit requests to generate images were compared to a control condition known to elicit images implicitly (Experiment 1). Other instructional manipulations were designed to alter the content of the images. Participants were asked to create generic or self-referential images of individual objects (Experiments 1 and 2) or integrated object sets (Experiments 2 and 3). Finally, the importance of source was examined by using participants' imagery descriptions as scripts for guiding other participants' imagery experiences (Experiment 3).

When prompting clients to engage in guided imagery (e.g. to explore fears or worries), therapists sometimes provide detailed scripts for guiding the imagined scenarios. And, during these guided imagery sessions, individuals may be encouraged to mentally transport themselves to places or situations, conjuring images of people, objects or events. However, as Arbuthnott and her colleagues point out, not all guided imagery techniques are this directive. On occasion, therapists provide just a few prompts, encouraging clients to use the prompts to construct an imagined scenario. Thus, with co-constructed imagery techniques, the client specifies the content of the script guiding the imagery generation.

Arbuthnott and her colleagues demonstrate that the source of imagery scripts seems to have consequences on the phenomenological characteristics of memories involving imagery experiences (Arbuthnott, 2005; Arbuthnott et al., 200X; Arbuthnott, Geelen, & Kealy, 2002; Kealy & Arbuthnott, 2003). Compared to memories for actual events, memories for imagery experiences based on others' scripts (Arbuthnott et al., 2002) are rated as lower in their sensory detail. In contrast, the phenomenological characteristics of memories for imagery experiences that are co-constructed, with the participant playing a role in the descriptions guiding the imagery experiences, are quite similar to those associated with memories for actual events (Kealy & Arbuthnott, 2003). These findings suggest that the characteristics of guided imagery techniques may have important

consequences for memory confusions. We examined this possibility in the present series of experiments by including both participant-generated imagery scripts (Experiments 1 and 2) and imagery scripts provided by someone else (Experiment 3). Instructional manipulations used to examine the influence of source and content of images were informed by the work of Arbutnott and her colleagues.

EXPERIMENT 1

The specific predictions tested in the first experiment in this series focused on the possible role of the nature of the imaginal process mediating the imagery experience. In Experiment 1, using a cover task shown to elicit implicit images (Foley et al., 1991), some participants were simply asked to describe functions for familiar objects (Function-only condition). Other participants were asked to create an image for each object, before describing a possible function for that object. Although objects were presented individually in all conditions, the objects were actually sets of related lists drawn from published norms. If requests to generate images function similar to the effects of pictorial presentations, then we would expect to find lower levels of false memory rates following explicit requests to generate images compared to the condition in which there was no mention of imagery.

To begin to examine the possible role of imagery content, participants were asked to create generic or self-referential images. In the one other investigation in which participants were asked explicitly to generate images for objects presented by way of a standard DRM procedure, there were no effects on false memory rates following requests to generate images, or requests to engage in self-referential processing in the absence of any explicit mention of images (Newstead & Newstead, 1998). Thus, one might doubt the likelihood that requests to generate images would have any effect on false memory rates. However, the absence of effects for these previous instructional manipulations relative to standard control conditions is difficult to interpret.

At least for some of the instances, the items on thematically related lists did not have corresponding discrete referents to image (e.g. doze and peace on the list related to sleep, bitter and sour on the list related to sweet). In the present series of experiments, like those examining the effects of pictorial encoding, the presented items were all highest associates that could be presented pictorially (and presumably could therefore be imagined). Given previous findings on the incidence of spontaneous self-referential processing (e.g. Foley et al., 1999), it is also at least plausible that participants in the earlier investigation were experiencing images (whether explicitly asked to do so or in response to requests to generate brief narratives), and these images may have been self-referential in nature at least on some trials. Therefore, we considered the possibility that self-referential processing might accompany imagery generations, in the absence of explicit requests, by asking participants to report on their images during encoding and, again, at test after their memory judgements were completed.

Method

Participants

Thirty undergraduates participated in Experiment 1, each receiving extra credit toward the completion of a course requirement for an introductory psychology course. Each undergraduate volunteer participated in only one experiment in this series.

Materials

Previous studies of modality effects on false memory rates, particularly those using pictorial presentations of associate lists (Israel & Schacter, 1997; Schacter et al., 1999), guided the construction of both the encoding and test series. As in these previous studies, our materials were created by adapting lists from published norms for inducing false memories in the DRM paradigm (e.g. Roediger & McDermott, 1995; Stadler, Roediger, & McDermott, 1999) as well as from the Russell and Jenkins (1954) word association norms. The eight highest associates that could be presented pictorially (Israel & Schacter, 1997), and presumably, could therefore be imagined, were selected from each of 16 associative lists. For each list (e.g. mug, saucer, tea, coaster, lid, coffee, straw, soup), an additional high associate was selected as the non-presented critical lure (e.g. cup). Ten of these 18 lists were presented in the encoding series, with the associates within each list presented in blocked fashion and in decreasing associative order. Although there were two different random presentation orders, the randomization was based on lists not items within the associate lists. The eight other lists were used to select distractor (or new) items for the memory tests.

The test construction procedures were also based on those used by researchers investigating the effects of pictorial encoding (e.g. Israel & Schacter, 1997; Schacter et al., 1999). A subset of the words presented during the encoding phase was included on the memory test. The 64-item memory test consisted of four kinds of test items: (1) 30 of the words presented during encoding, referred to here as *presented instances*, were included by selecting the presented items in each list that occurred in the 1st, 3rd and 7th positions in the lists (or blocks); (2) 10 non-presented critical items, each a high associate of the words presented in lists, referred to here as *critical lures*; (3) Eighteen new words, referred to here as *new category instances*, three from each of six lists not presented during encoding, and again, chosen from the 1st, 3rd and 7th positions of their respective associative lists; and (4) Six new high associates, one from each of the six associative lists not presented, and referred to here as *control lures*. The items on the recognition tests were randomly presented with at least one instance presented from each of the 18 associate lists before a second instance of any list was presented. Thus, although item presentations were blocked during the encoding phase, they were not blocked during the test phase.

Samples of Bev Doolittle's art were also included as materials for a search task serving as a distractor during the retention interval. These art selections portrayed several complex scenes with many camouflaged objects (e.g. faces, horses) within each. The objects on the search tasks embedded in these scenes were not referred to during the encoding or test phase of the experiment.

Procedure

Participants were randomly assigned to one of the three instructional encoding conditions, two of which included explicit requests to generate images. Across the three instruction conditions, each encoding trial consisted of the single presentation of a word with the experimenter reading the words and recording participants' descriptions of objects' functions. Each of these encoding trials lasted approximately 5 seconds. Participants were given the impression that we were collecting norms about objects' functions to be used in selecting materials for developmental studies. This cover task effectively focuses attention on the materials without making reference to a memory test (e.g. Foley et al., 1991).

In the Function-only Condition, intended as an implicit imagery control condition, participants were simply asked to describe a function for each object (i. e. 'describe

something that might be done with the object'). In the Generic Imagery condition, participants were asked to 'generate an image for each object, and once the image comes fully to mind, describe something that might be done with that object.' In the Self-Referential Imagery condition, again, participants were asked to generate an image of each object and then to describe a function for the object. However, they were also asked 'to create an image for each object that connects in some way to your life, perhaps visualizing an object of your own or thinking about an episode from your life involving the use of the object. Once the image comes to mind fully, describe something that you might do with the object.'

After a 7-minute retention interval, during which time participants searched for camouflaged objects embedded in complex scenes, participants were surprised with the 64-item old/new recognition test. Listening as the experimenter read the 64-item test list, one at a time, participants were asked to report whether or not each test item was one presented during the first phase of the experiment. After the test phase, participants in all conditions were asked whether they expected a memory test. In addition, participants in the two explicit imagery instructional conditions were asked what helped them remember the words that were presented during the encoding phase. When the experiment was completed, participants were fully debriefed.

Results

Dependent measures and overall approach to data analyses for Experiments 1–3

In all experiments reported in this series, old/new recognition tests were scored to construct four dependent variables: (1) Proportion of correct 'yes' responses to words presented during the encoding phase; (2) Proportion of incorrect 'yes' responses to non-presented words related to words that were presented, referred to here as *critical lures*; (3) Proportion of incorrect 'yes' responses to high associates of each of the semantic categories from which the new instances were drawn, referred to here as *control lures*; and (4) Proportion of incorrect 'yes' responses to instances of semantic categories not represented in the encoding series, referred to here as *new instances*. Note that *false memories* in the DRM paradigm refer to critical lures.

For each experiment, one Analysis of Variance (ANOVA) was computed on proportion correct 'yes' responses. A second ANOVA was computed on incorrect 'yes' responses, including type of error as a repeated-measures variable. The three kinds of errors were included in one analysis to allow for tests of predictions about the effects of imaginal encoding on critical lures (or false memory errors), and to compare error rates against their intended controls. When the Analyses of Variance (ANOVA) revealed significant effects, if the effects were based on a between-groups variable, Tukey-Kramer *post hoc* tests were then conducted. If the effects were based on a repeated-measures variable, then Scheffe's test was used to correct the critical F-value (Keppel & Wickens, 2004). The 0.05 level of significance was used for all analyses.

Proportion correct 'yes' responses

Table 1 reports both correct and incorrect 'yes' responses as a function of the Imagery Instruction conditions. Analysis of proportion correct 'yes' responses revealed a main effect of instructional manipulation, $F(2, 27) = 14.61$, $MSE = 0.01$, $p < 0.001$. *Post hoc* tests showed that recognition of presented items was quite good and comparable in the two explicit imagery instruction conditions (Generic and Self-Referential), but significantly

Table 1. Proportion correct yes responses to presented items and proportion incorrect yes responses to critical lures, new instances and control lures, as a function of instruction condition, Experiment 1

	Function instruction condition					
	Function only		Generic imagery		Self referential Imagery	
	<i>M</i>	<i>SE</i>	<i>M</i>	<i>SE</i>	<i>M</i>	<i>SE</i>
Proportion correct yes to						
Presented items	0.78	0.03	0.95	0.02	0.94	0.02
Proportion incorrect yes to						
Critical lures	0.61	0.02	0.10	0.03	0.23	0.04
Control lures	0.09	0.03	0.12	0.03	0.14	0.03
New instances	0.07	0.02	0.00	—	0.00	—

higher in these two conditions than in the Function-only condition (with the critical mean difference = 0.09).

Proportion incorrect 'yes' responses to three sets of new items

A 3(Instruction Type) $\times$ 3(Type of Error) ANOVA on incorrect 'yes' responses revealed three significant effects: A main effect for Instruction, $F(2, 27) = 55.29$, $MSE = 0.01$, $p < 0.001$, a main effect for Type of Error, $F(2, 54) = 96.02$, $MSE = 0.01$, $p < 0.001$, and a significant Instruction $\times$ Error interaction, $F(4, 54) = 33.31$, $MSE = 0.01$, $p < 0.001$.

As predicted, and confirmed by post hoc tests, incorrect 'yes' responses to critical lures (or false memories) were greater in the Function-only condition ($M = 0.61$) than in the Generic ($M = 0.10$) or Self-Referential Imagery ($M = 0.23$) conditions. Incorrect 'yes' responses to critical lures were also greater in the Self-Referential than Generic Imagery condition. The critical mean difference for all three comparisons was 0.11.

Error rates on control lures did not differ across the three instruction conditions. Finally, although error rates on new instances were quite low, these rates were higher in the Function-only condition ($M = 0.07$) than in either the Generic or Self-Referential Imagery Conditions ($M_s = 0.00$ and 0.00 , respectively, (with critical mean difference = 0.03).

When analyzing error rates within the Function-only condition, the simple main effect for error type was significant, $F(2, 18) = 213.42$, $F_{crit} = 8.69$. Individual mean comparisons showed that error rates on critical lures ($M = 0.61$) were higher than error rates on control lures ($M = 0.09$) and new instances ($M = 0.07$), $F_s = 209.78$ and 656.10 , respectively, with $F_{crit} = 8.69$. The latter two means did not differ. Similarly, In the Self-Referential Imagery condition, the simple main effect for error type was significant, $F(2, 18) = 12.00$. Individual mean comparisons showed that the error rates on critical lures were higher ($M = 0.23$) than the error rates on control lures ($M = 0.12$) and new instances ($M = 0.00$), $F_s = 16.39$ and 29.57 , respectively. The latter two means did not differ. Finally, in the Generic Imagery condition, the simple effect for error type was not significant.

Content analysis on encoding descriptions

Participants' references to critical lures. When generating descriptions for objects' functions (e.g. a table), participants may have mentioned one of the non-presented critical lures in their descriptions (e.g. a chair), possibly contributing to subsequent memory

confusions. The number of times the critical lures were mentioned in participants' descriptions was actually quite low (on an average 1 out of 10 possible times), and this incidence did not vary across conditions.

Coding for self-referential remarks. In all conditions in Experiment 1, descriptions of objects' functions were also coded for the presence of self-referential remarks. Two research assistants unfamiliar with the purpose of the study, coded descriptions for the number of times (out of 80 word presentations) that participants made explicit reference to their own objects (e.g., 'my desk') or to some contextual setting from their lives (e.g. 'the desk in my office'). The incidence of explicit self-referential remarks was quite low across the three encoding conditions, on average, on less than 10% of the encoding trials. Because participants' descriptions of objects' functions were relatively short—e.g. including phrases like 'drinking from it' (for mug), 'closing it' (for door) or 'spreading it' (for butter), the encoding descriptions may have underestimated the frequency of self-referential processing.

The absence of explicit self-referential remarks in the descriptions does not preclude the possibility that participants might have experienced (or created) images that were self-referential in nature. Consistent with this possibility, following the completion of the memory test, when participants were asked how they decided whether or not a test item was presented previously, many of the participants in both imagery conditions made reference to thinking about the images they created, e.g. reporting: 'it was easy to create images related to people or personal things like my dog or our dining room table at home, and these images came to mind when I heard the words on the test'. The incidence of these kinds of self-referential remarks was relatively frequent in the two generate conditions (on 80% or more of the trials) but was rare in the Function-only condition with most participants reporting: 'it's hard to say' (or the equivalent) when asked to describe the basis for their old/new recognition judgements.

Discussion

Experiment 1 demonstrates that the processes giving rise to imagery experiences (spontaneous vs. deliberate) play a role in the expression of false memory rates. Generating images of familiar objects significantly reduced false memory errors relative to an imagery control condition (Function-only condition) in which explicit requests to generate images were not mentioned. Generic Imagery instructions led to lower false memory rates than Self-Referential Imagery instructions, suggesting that the content of the images may also play a role. To examine further the possible effects of image content, in Experiment 2, we replicated the imagery instruction conditions of Experiment 1, asking participants to create generic or self-referential images, but lengthened the encoding trials. We also introduced a second instructional manipulation, asking participants to generate images of individual objects or integrated object sets.

According to the impoverished-encoding explanation, picture presentations (and by extension requests to generate images) are thought to induce item-specific processing, and thereby reduce the likelihood of relational processing giving rise to the activation of implicit associate responses. If requests to generate images of individual objects were more likely to induce item-specific processing, and thereby minimize relational encoding, then we would expect false memory rates to be lower when generating individual images compared to integrated ones. Moreover, requests to generate integrated images may

enhance relational encoding, thereby leading to increases in false memories at test. Similar predictions follow from the distinctiveness heuristic. If requests to generate integrated images are more likely to induce implicit images of related but non-presented items (relative to requests to generate individual images), then at test, monitoring presented from non-presented related items should be more difficult leading to an increase in false memories.

In all four imagery conditions defining Experiment 2, participants were explicitly asked to generate images of familiar objects before describing object functions. The experimenter read the thematically related words lists, one word at a time or in subsets of four related words, and participants were asked to generate images of individual objects or integrated object sets. Thus, in Experiment 2, instructional manipulations varied in two ways, focusing on the number of objects included on each encoding trial (Individual vs. Integrated object sets) as well as the specificity of the imagery instructions (Generic vs. Self-Referential).

EXPERIMENT 2

Method

Participants

Forty-eight undergraduates participated in Experiment 2, each receiving extra credit toward the completion of a course requirement for an introductory psychology course.

Materials

The materials were identical to those included in Experiment 1. From participants' perspectives, on each encoding trial, items were presented individually (with the experimenter reading one word at a time, e.g. table) or in sets of four related objects (with the experimenter reading four related words together, for example, table, couch, desk, lamp). From the experimenter's perspective, for both types of encoding trials, the presentation order preserved the word associate relationships with the instances in each list presented in descending associative order as in Experiment 1. For the integrated imagery trials, the ordering of the eight items in each list was preserved but two subsets of four items were presented for the integrated versions. Thus, there were 80 individual encoding trials in the individual imagery conditions but 20 encoding trials in the integrated imagery conditions, each involving a 4-item set. As a result, the overall encoding phase was somewhat longer for individual-item trials. However, on all encoding trials, participants typically completed their imagery generation and function descriptions within 10 seconds of the initial presentation of the material (single word vs. set of four words).

As in Experiment 1, there were also two different randomized encoding orders but these randomized orders did not disrupt the associative sets. The items on the old/new recognition test were also identical to those used in Experiment 1. Test items were presented randomly with at least one instance from each of the 18 associate lists presented before a second instance of any associate list was presented. The search task materials described for Experiment 1 were again included in this second experiment.

Procedure

In all conditions, participants provided descriptions of the images that came to mind, identifying object functions, and an experimenter recorded these descriptions. In the

Individual Imagery conditions, each trial consisted of a single word presentation whereas in the Integrated Imagery conditions a single trial consisted of four words read out loud. After a 7-minute retention interval, during which time the participants searched for camouflaged objects embedded in complex scenes, they were surprised with the 64-item old/new recognition test used in Experiment 1. Test items were read and presented individually.

Before the target items were presented, two sample trials were included in each of the four imagery conditions. To illustrate the instructions for all four conditions, one of these examples is included here. When a practice item such as *sand pail* was presented alone, participants were instructed to 'imagine a sand pail, and once the image comes to mind fully, describe something that might be done with a sand pail' (Generic Imagery Condition) or 'to generate an image of a sand pail that connects in some way to your life, perhaps visualizing a sand pail of your own or an episode in your life that involved (or could involve) the use of a sand pail. Once the image comes to mind fully, describe something that you might do with a sand pail' (Self-Referential Imagery Condition).

By comparison, when the item *sand pail* was included with three other related items (*shovel*, *sand* and *box*) to illustrate a four-item trial, participants in the Generic/Integrated Imagery condition were told 'to create an image involving a sand pail, a shovel, sand and a box. Once the image comes to mind fully, describe something that might be done with the objects in the image'. Participants in the Self-Referential/Integrated Imagery condition were asked to 'create an image involving a sand pail, a shovel, sand, and a box. Generate an image of these objects that connects in some way to your life, perhaps thinking about your own versions of these objects or an episode in your life that involved (or could involve) using them. Once the image comes to mind fully, describe something that you might do (or have done) with these objects in the image'.

Results

Proportion of yes responses to item sets

Table 2 reports both correct and incorrect 'yes' responses as a function of the two imagery instruction manipulations.¹ An Analysis of Variance (ANOVA) including Type of instruction (Generic vs. Self-referential) and Type of image (Individual vs. Integrated Object Images) on the proportion correct 'yes': responses to presented items revealed one significant effect'. Accuracy for presented items was higher after generating individual

¹Analyses on the content of the imagery descriptions for Experiment 2 were similar to those reported for Experiment 1. Therefore, they were not reported in detail here except to mention that the incidence of explicit self-referential remarks in the two integrated imagery conditions (at least 50% of the trials) was greater than that for the individual imagery conditions. Because these descriptions were also relatively longer, the opportunity for these self-referential remarks was greater. Participants might have responded to the integrated imagery instructions by simply visualizing four objects at the same time, but not integrated in any way. Coding the participant-generated descriptions for evidence of integrated responses indicated that on at least half of the trials (twenty 4-item trials), participants made explicit reference to the integrated nature of their visualizations (e.g. describing an occasion in which all four objects might have been used to accomplish some goal). Participants did not always mention all four objects in their descriptions, repeating back the set listed by the experimenter. Our instructions did not require participants to mention all four objects, in part, because we were intent on avoiding any specific guidance about imagery content other than the words in each set. Furthermore, we did not require participants to repeat back the items in the same order as we presented them. When looking at the descriptions for the number of items in each set that were explicitly mentioned by the participants, most participants (85%) mentioned at least three of the four items in their descriptions.

Table 2. Proportion correct yes responses to presented items and proportion incorrect yes responses to critical lures, new instances, and control lures, as a function of imagery instruction and type of image, Experiment 2

	Type of image							
	Individual				Integrated sets			
	Imagery instruction							
	Generic		Self referential		Generic		Self referential	
	<i>M</i>	<i>SE</i>	<i>M</i>	<i>SE</i>	<i>M</i>	<i>SE</i>	<i>M</i>	<i>SE</i>
Proportion correct yes to:								
Presented items	0.97	0.01	0.98	0.01	0.86	0.01	0.86	0.03
Proportion incorrect yes to								
Critical lures	0.32	0.04	0.17	0.04	0.47	0.07	0.44	0.07
Control lures	0.03	0.02	0.00	—	0.07	0.04	0.07	0.04
New instances	0.00	—	0.00	—	0.03	0.01	0.03	0.02

images ($M=0.97$) than integrated ones ($M=0.86$), $F(1, 44)=54.47$, $MSE=0.003$, $p<0.001$. This effect did not interact with the other imagery variable.

Proportion of incorrect 'yes' responses to three item types

In an ANOVA including Type of instruction (Generic vs. Self-referential), Type of image (Individual vs. Integrated Object Images), and Type of Error (on critical lure, control lure, new instance) on the proportion of incorrect 'yes' responses, three effects were significant. There was a main effect for Type of Image, $F(1, 44)=12.77$, $MSE=0.03$, $p<0.001$, a main effect for Type of Error, $F(2, 88)=145.43$, $MSE=0.01$, $p<0.001$ and a significant Image $\times$ Error interaction, $F(2, 88)=9.75$, $MSE=0.01$, $p<0.001$. To locate the basis of the interaction, *post hoc* tests were conducted.

Post hoc tests indicated that incorrect 'yes' response to critical lures (or false memories) were greater after generating integrated object images ($M=0.45$) than individual ones ($M=0.25$), critical mean difference = 0.12. However, incorrect responses to control lures did not differ for integrated and individual imagery generation conditions. Although errors on new instances were quite low, these errors were higher following the generation of integrated object images ($M=0.03$) than individual ones ($M=0.00$), critical mean difference = 0.03.

When analyzing error rates within the Single Imagery conditions, the simple main effect for error type on incorrect 'yes' responses was significant, $F(2,46)=52.13$, $F_{crit}=6.37$. Individual mean comparisons showed that error rates on critical lures ($M=0.25$) were higher than the error rates on control lures ($M=0.01$) and new instances ($M=0.00$), $F_s=47.82$ and 61.82 , respectively. The *post hoc* test comparing the latter two means was not significant. Similarly, when analyzing error rates within the Integrated Imagery conditions, the simple main effect for error type was significant, $F(2,46)=89.99$, $F_{crit}=6.37$. Individual mean comparisons showed that errors rates on critical lures ($M=0.45$) were higher than error rates on control lures ($M=0.70$) and on new instances ($M=0.03$), $F_s=103.28$ and 100.70 , respectively. The *post hoc* test comparing the latter two means was not significant.

Although the interaction between Type of imagery instruction (Generic vs. Self referential) and Type of Error on incorrect 'yes' responses was not significant, because two imagery conditions in Experiment 2 were procedurally similar to the Generate Imagery conditions of Experiment 1, the incidence of incorrect 'yes' responses to critical lures in the Generic and Self Referential Individual Imagery conditions was compared. *Post hoc* tests showed that the errors rates on critical lures were higher in the Generic ($M = 0.32$) than in the Self Referential ($M = 0.17$) condition with mean critical difference = 0.12. The direction of this difference is inconsistent with that observed for Experiment 1. By way of reminder, in Experiment 1, the means were .10 and .23 for the Generic and Self-referential conditions, respectively. And, for completeness sake, this inconsistent pattern was specific to errors on critical lures; overall recognition of target items was similar for comparable conditions in Experiments 1 and 2.

Discussion

The results of Experiment 2 demonstrate two new findings. Generating individual images led to better recognition of presented items and lower false memory rates, consistent with the impoverished encoding explanation. In much the same way that seeing pictures of objects during encoding is thought to support item-specific processing, imagining objects individually with self-generated images may lead to item-specific processing, resulting in more distinctive encoding, and leading to memory enhancement effects (e.g. Hunt & McDaniel, 1993). As mentioned earlier, the length of the encoding phases differed for the Individual and Integrated imagery conditions with relatively longer encoding trials in the Individual imagery conditions. This difference in encoding time could also have contributed to higher accuracy rates (on both presented items and critical lures).

Although imagining objects individually with self-generated images resulted in the lowest false memory rates (Experiment 2), whether images of individual objects were self-referential or generic had inconsistent effects. Compared to the Self-referential condition, false memories were lower in the Generic Imagery condition of Experiment 1 but greater in the Generic/Individual Imagery condition of Experiment 2. Why the enhancement effects following the generation of images of individual objects were inconsistent awaits further study. The one procedural difference between Experiments 1 and 2 suggests the possibility that the length of the encoding phase may have played a role by affecting the opportunity for relational processing.

The length of the encoding phase was longer in Experiment 2 (10 sec) than Experiment 1 (5 sec). When creating individual object images under generic imagery instructions, longer encoding trials could increase the likelihood of implicit activations (e.g. semantic associates or images of related objects) leading to greater false memory rates. Consistent with this reasoning, comparing across Experiments 1 and 2, the relative differences in false memory rates were greater for the generic than self-referential imagery conditions. If self-referential encoding supports item-specific processing, a similar increase in the length of the encoding trials would not be expected to influence (at least to the same degree) the implicit activation of semantic associates.

Particularly because the relationship between encoding duration and false memory rates is a complex one (e.g. McDermott & Watson, 2001), an explanation for the inconsistent effects observed within the individual imagery conditions of Experiments 1 and 2 await further study. The crucial point for present purposes is that even with this inconsistency, the level of false memory rates are considerably reduced relative to an imagery control

condition (Experiment 1) and to rates reported for standard versions of the DRM paradigm. In future studies, measuring the time to actually generate the images may be informative when assessing the basis for the effects of image content. Because the time invested in generating images may differ for various guided imagery techniques, pursuing the ways in which encoding duration may affect the nature of elaborative processing may be particularly important for applied settings.

EXPERIMENT 3

Critics of the use of guided imagery techniques argue that spontaneous thoughts prompted by guided imagery may take on a life of their own, with clients increasingly convinced that these thoughts emanate from actual experiences. Nevertheless, variations in guided imagery techniques are rarely taken into account in any systematic way when criticisms are levied against their use. In the memory literature, we do find variations in guided imagery instructions. Studies of the imagination inflation effect typically involve the use of experimenter-defined scripts specifying relatively elaborate imagined scenarios (e.g. Garry et al., 1996; Hyman & Pentland, 1996). By comparison, studies of the negative effects on memory of non-repeated imagination trials include both experimenter-defined (Hashtroudi et al., 1990) and co-constructed (e.g. Foley et al., 1989) imagery scripts. However, investigations of the effects of imagery source (participant or someone else) on memory confusions have not been conducted.

The primary purpose of Experiment 3 was to see if shifting the source of the scripts to someone other than the participants would have consequences for false memory rates. If we show that the degree of memory confusion about what was actually presented or was only experienced in thought depends on the source of the imagery script, our results would emphasize the importance of paying careful attention to the features of guided imagery techniques when arguing against their use in applied settings. In Experiments 1 and 2, imagery scripts were co-constructed in that the experimenter provided prompts (individual or integrated sets of objects) and participants then used these prompts to guide their own imagery experiences. In contrast, in Experiment 3, the imagery instructions included brief scripts specified by others (i.e. participants in self-generated conditions).

Although theoretical explanations for false memory errors induced by the DRM procedure often conceptualize these errors as monitoring failures, the error rates may reflect limits on response options (e.g. Multhaup & Conner, 2002). Under standard DRM Old/New recognition testing conditions, participants are instructed to report old only to those items that they think were actually presented. Thus, from the experimenter's perspective, these instructions imply that items experienced only in thought should not be classified as old. However, this assumption may not be as obvious to participants. Therefore, in Experiment 3, we included the opportunity to report 'experienced in thought only' by including two types of test, a standard Old/New recognition test and a Heard/Thought Source test.

Method

Participants

The 24 undergraduates who participated in this experiment received extra credit toward the completion of a course requirement for an introductory psychology course.

Materials

Words were presented in sets of four, paralleling the integrated imagery conditions of Experiment 2. These word sets (and test items) were identical to the materials included in the first two experiments in this series. In the last experiment, however, when participants were instructed to generate images of objects, requests to generate images were accompanied by scripts for guiding participants' visualizations. Specifically, we randomly selected 12 sets of descriptions from those generated under integrated imagery instructions as materials for Experiment 3. When creating these scripts, self-referential remarks (e.g. 'my desk' 'my mother's') including references to specific occasions (e.g. 'my grandmother's birthday party') were deleted from participant-generated descriptions. Nevertheless, the scripts preserved the episodic nature of the descriptions where appropriate (e.g. reference to a relative's birthday party was retained). If a participant's description included any re-ordering of the items mentioned in a subset (e.g. referring to box, shovel, sand pail and sand in response to the list 'sand pail, shovel, sand, and box'), the imagery script that the experimenter read preserved that reordering. Table 3 provides examples of the participant-generated descriptions and scripts derived from these descriptions.

Procedure

At the start of each encoding trial, the experimenter read a subset of four words in the same associative order as that used for Experiments 1 and 2. Immediately after reading each subset of four words, the experimenter then read a brief script, asking participants to use that script to guide their imagery generation. After generating an image based on the script, participants were whether each description resembled something they might do with the objects mentioned. Each encoding trial (defined by the presentation of a four-word set, the script referring to the words in the set, and the participant's response) lasted approximately 10 seconds. Participants typically reported that the descriptions (and corresponding

Table 3. Illustration of three kinds of imagery encoding descriptions

Word Set: *Table, Couch, Desk and Lamp*

Participant-Generated Imagery Descriptions

Participant A:

'Sitting on my couch, picturing using a lamp on a table to read a book that's resting on a desk'

Participant B:

'I am thinking of my living room at home, picturing the table with magazines on it, I'm sitting on the couch and using a lamp on the desk there to read'

Participant C:

'I was just with my family and I'm thinking of that time with us sitting around the living room, I'm picturing our table with lots of magazines, we're sitting on the couch looking at the magazines, turning on a lamp on the desk so we can see better'

Script-Based Imagery Descriptions

Yoked Participant A:

'Imagine yourself sitting on your couch, using a lamp on a table to read a book that is resting on a desk'

Yoked Participant B:

'Imagine yourself in a living room with a table holding magazines. You're sitting on a couch and using a lamp to read a book that is set on a desk nearby'

Yoked Participant C:

Imagine a time you were home with your family sitting around the living room, picture a table with lots of magazines on it, sitting on the couch, and turning on a lamp near a desk so you could read.

images) were similar to actions they might actually perform with the objects mentioned. A small percentage reported 'no,' but only on a small number of the trials. After a 7-minute interval during which time participants worked on the search tasks used in Experiments 1–2, they were surprised by an Old/New Recognition test or a Hear/Think Source test.

For the standard Old/New Recognition test, participants were asked to decide whether or not each test item was presented earlier as part of the encoding phase. For the Heard/Thought Source Test, participants were asked to decide whether each test item was a word that they heard the experimenter read during the encoding phase, a word they remembered thinking about but did not hear presented when listening to the imagery descriptions, or a new word not included in the encoding series.

Results

Proportion correct 'yes' responses

Table 4 summarizes correct and incorrect 'yes' responses as a function of Type of memory test. An ANOVA including Type of Test on correct responses revealed no significant effects. Accuracy levels resemble those reported for Experiment 2 (see Tables 2 and 4).

Proportion incorrect 'yes' responses

When calculating the number of incorrect yes responses to lures in the Heard/Thought Source Test condition, incorrect responses were defined as (false) claims that these words were 'heard' during the encoding phase. An ANOVA including Type of Test and Type of Error on the number of incorrect 'yes' responses revealed a main effect for Type of Error, $F(2, 44) = 160.15$, $MSE = 0.01$, $p < 0.001$. The simple main effect for type of error was also significant, $F = 158.20$, $F_{crit} = 6.60$. Subsequent *post hoc* tests showed that there were more errors on critical lures ($M = 0.58$) than on the control lures ($M = 0.07$) and than on new instances ($M = 0.03$), $F_s = 143.79$ and 336.42 .

Discussion

Because the materials on which images were based were comparable across Experiments 2 and 3, with only the source of the images differing, relative comparisons between accuracy

Table 4. Proportion correct yes responses to presented items and proportion incorrect yes responses to critical lures, new instances, and control lures, as a function of type of memory test, experiment 3 (imagery scripts specified for participants)

	Type of memory test			
	Old/new recognition		Heard/thought source test	
	<i>M</i>	<i>SE</i>	<i>M</i>	<i>SE</i>
Proportion correct yes to:				
Presented items	0.85	0.02	0.87	0.02
Proportion incorrect yes to:				
Critical lures	0.55	0.04	0.62	0.04
Control lures	0.12	0.05	0.07	0.04
New instances	0.03	0.01	0.03	0.01

on presented items and false memory rates on critical lures are appropriate and informative. Accuracy levels were quite similar for the integrated imagery conditions of Experiments 2 and 3. However, false memory rates differed considerably (on average, 45 and 58% for Experiments 2 and 3, respectively). Although false memory rates did not differ for the two testing conditions in Experiment 3, it may be premature to conclude that false memory rates are insensitive to options provided at test. Previous investigations of the effects of source tests lead to conflicting findings, with some reports of an increase in false memory rates (e.g. Hicks & Marsh, 2001) and others reporting a decrease (Multhaup & Conner, 2002). The inconsistent effects of source tests invite closer examination of the nature of the options provided at test (i.e. whether these options focus on the materials themselves—e.g. voice of the speaker presenting the materials—or the kinds of processing that accompany the processing of materials—e.g. heard vs. self-generated).

GENERAL DISCUSSION

The present series of experiments contribute several new findings to the literature by demonstrating the effects of imaginal encoding on false memory rates in the DRM paradigm. These imaginal encoding effects resemble those reported previously following pictorial encoding in that both kinds of encoding result in relatively lower rates of false memories than those reported following standard verbal presentation of materials in the DRM paradigm—overall 31% in Experiment 1 and 35% in Experiment 2 following imaginal encoding, 30% following pictorial encoding (e.g. Israel & Schacter, 1997, Experiment 2), and 70–80% under standard encoding conditions (Roediger et al., 1998). Nonetheless, the processes giving rise to imaginal encoding (Experiment 1) as well as its content (Experiment 2) and source (Experiments 2 and 3) have consequences for these false memory rates. False memory rates were relatively higher when images were experienced spontaneously (Experiment 1), when images involved integrated sets (Experiment 2), and when participants were given scripts to guide their imagery experiences (Experiment 3).

Implications for explanations of pictorial encoding effects in the DRM paradigm

Collectively our findings are consistent with the distinctiveness heuristic in its prediction that false memories should decrease following the deliberate generation of images. When images (or memories for pictures) serve as a diagnostic, monitoring decisions may focus on the presence (or absence of images) or on the operations giving rise to those images. If images for presented items are re-experienced at test, participants may use the absence of an imagery experience as an indication that the item was not presented earlier in the session. Generating images during encoding may also lead to the activation of implicit images for related materials, thus, both presented and non-presented related test items may evoke images at test. Successful monitoring decisions in this instance could reflect the use of cues based on cognitive operations information (e.g. remembering the experience of choosing and generating a particular image for presented items).

Generating integrated images led to more false memories than generating individual ones, consistent with predictions derived from both the *impoverished encoding* and *distinctiveness heuristic* hypotheses. Although the present studies were not intended as a test of the independent role of activation and monitoring processes, their results do suggest

that the distinctiveness heuristic is sensitive to features of the images themselves, and, hence the nature of the imaginal encoding plays a crucial role in the incidence of false memory errors. If the distinctiveness heuristic were based solely on general assumptions about the memorable quality of sets of materials (whether experienced as images or pictures), or on global impressions of the distinctiveness of encoding lists, we would have expected to observe comparable levels of false memories across variations in instructions about the content of images.

The increase in false recollections following the generation of integrated images could be used to lend further support to concerns about the use of guided imagery. However, other aspects of our findings suggest that drawing this implication is unwarranted. For integrated images, the incidence of false recollections is considerably lower when the images are specified by participants (Experiment 2) rather than provided by others (Experiment 3). Thus, when evaluating the implications of the use of guided imagery on memory confusions, the source of the image (participant or other specified) may be at least as important as the integrated nature of the imagined scene.

Logically, implicit responses could be evoked during encoding, at test with the presentation of related items (Roediger & McDermott, 1995), or in response to questions about the memories themselves (e.g. Gallo & Roediger, 2003). Recent research suggests that activations that occur during encoding play a key role in these memory errors under standard DRM conditions (Marsh, McDermott, & Roediger, 2004). Whether this is also the case under conditions of imaginal processing remains to be determined.

Implications for debates about guided imagery techniques

According to the Source-Monitoring framework motivating our work, thoughts may be mistaken for actual encounters when those thoughts bear some resemblance to the qualities of perceived events. Thus, when memories for thoughts (e.g. anticipations, images or intentions) are rich in detail, participants may later leap to the conclusion that they are remembering actual events. Along similar lines, in clinical settings, if individuals are guided to construct imagined scenarios rich in their perceptual detail, emotional power or semantic associations, it would not be surprising that these individuals might come to believe the imagined scenarios were actually experienced. Some researchers have suggested that most guided imagery techniques are similar to procedures used to study false memory rates in that both serve to lower the client's (or participant's) criterion for judging false recollections (including memories for thoughts) as veridical memories. However, as we have shown in the present series of experiments, the extent to which participants are confused about what they experienced directly and what they experienced only in thought very much depends on the features of the imagined experience.

Admittedly, there are important differences between the use of guided imagery techniques in applied and laboratory settings. Nevertheless, guided imagery techniques and our imagery instructions share a number of features. When using guided imagery, participants (or clients) may be invited to participate in the co-construction of the images. Alternatively, they may be given detailed scripts provided by others on which to base those images. The instructions may direct attention to previous experiences (asking participants or clients to describe perceptual details, feelings and thoughts that accompany the re-encountering of these experiences). Or, instructions may focus attention on a hypothetical event without reference to the individuals' previous life experiences. In Experiment 2, participants' imagery descriptions suggested that, at least on occasion, they spontaneously

contextualized their images, recollecting episodes from their lives (see Table 3, Participant C's description as an example). The nature of the focus (actual or hypothetical) may also prove to be of importance when predicting the effects of guided imagery on memory confusions. More generally, the study of relational processing in the context of guided imagery is a particularly intriguing direction for future work. In more naturalistic contexts, thematic relationships among seemingly unrelated items might exist by way of autobiographical experiences. Thus, asking participants to generate integrated images of unrelated item sets (e.g. mirror, pencil, stone and feather) could evoke relational processing by way of recollections of episodes involving those objects (e.g. a drawing class).

In conclusion, voices fueling debates about the causes and significance of memory confusions in general, and false memories in particular, often emphasize our susceptibility to the powers of the imagination. Indeed negative effects of imagination on remembering lead many to argue that we are at the mercy of our own imaginal processes. However, the present series of studies suggests caution in assuming that the deleterious effects of imagery are independent of the way in which images are induced or encouraged. Most notably, the source of the imagery script (self or other person) affects reports of false memories, leading to reductions in error when participants specify the content of those images. Thus, careful attention should be paid to the degree to which clients (or participants in studies of memory confusion) are directing their own guided imagery experiences. Until more is known about the effects of various kinds of guided imagery techniques on memory, it seems premature to leap forward with definitive suggestions about their use (or discontinuation).

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